

Computational Physics Classes No.2 - Report

Simulation of mechanical resonance for one-dimensional harmonic oscillator with RK4 method

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Abstract

This report presents a numerical study of a one-dimensional harmonic oscillator using the fourth-order Runge–Kutta (RK4) method. The simulations include undamped, damped, and driven cases to analyze energy conservation, dissipation, and mechanical resonance. The results show excellent agreement with analytical solutions and demonstrate how damping reduces oscillation amplitude and broadens the resonance peak.

1 Introduction

In this report, we investigate the dynamics of a one-dimensional harmonic oscillator subjected to dissipative and external driving forces. The motion of the particle is governed by the second-order differential equation derived from Newton’s law, which includes the restoring force, the damping term proportional to velocity, and an optional periodic external force. Analytical solutions exist only for simple cases, therefore numerical integration methods are required to study more complex situations. The Runge–Kutta method of fourth order (RK4) is implemented to obtain accurate solutions for the oscillator’s motion and to analyze phenomena such as energy conservation, damping effects, and mechanical resonance.

2 Description of the Numerical Method

The equation of motion of the oscillator is transformed into a system of two first-order differential equations for position and velocity. This system is solved numerically using the fourth-order Runge–Kutta (RK4) method, which provides high accuracy for ordinary differential equations. At each time step, four intermediate slope evaluations are performed to compute the next values of position and velocity. The algorithm is implemented in its vector form, allowing for efficient and clear numerical computation of the oscillator’s trajectory and related physical quantities such as kinetic, potential, and total energy.

3 Results

3.1 Undamped and Damped Oscillator ($\alpha = 0$ and $\alpha = 0.1$)

In the first simulation, the friction and external driving forces were disabled ($\alpha = 0$, $F_{\text{ext}} = 0$). Figure 1 shows the time evolution of position and velocity. The corresponding phase-space trajectory in Figure 2 forms a perfect closed loop, confirming that the total energy of the system is conserved. This is further illustrated in Figure 3, where the kinetic and potential energy oscillate periodically while their sum, E_{tot} , remains constant.

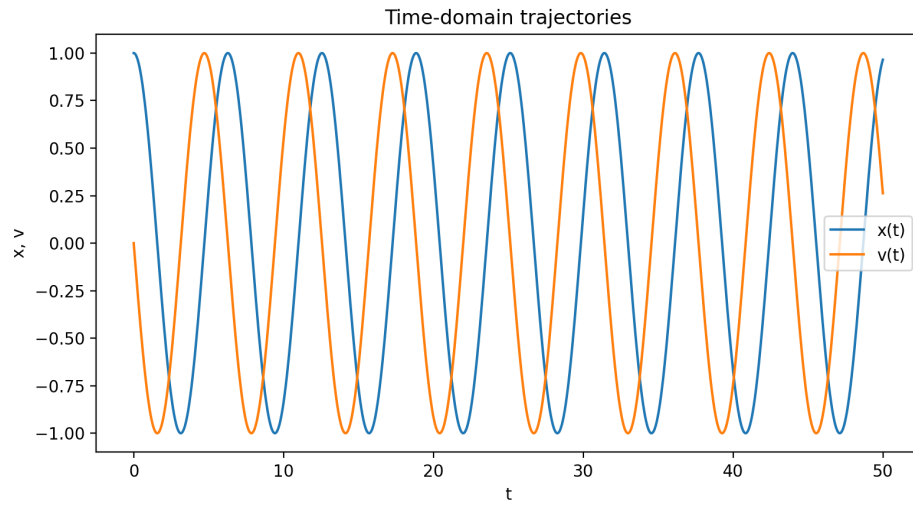


Figure 1: Time-domain trajectories $x(t)$ and $v(t)$ for the undamped oscillator ($\alpha = 0$).

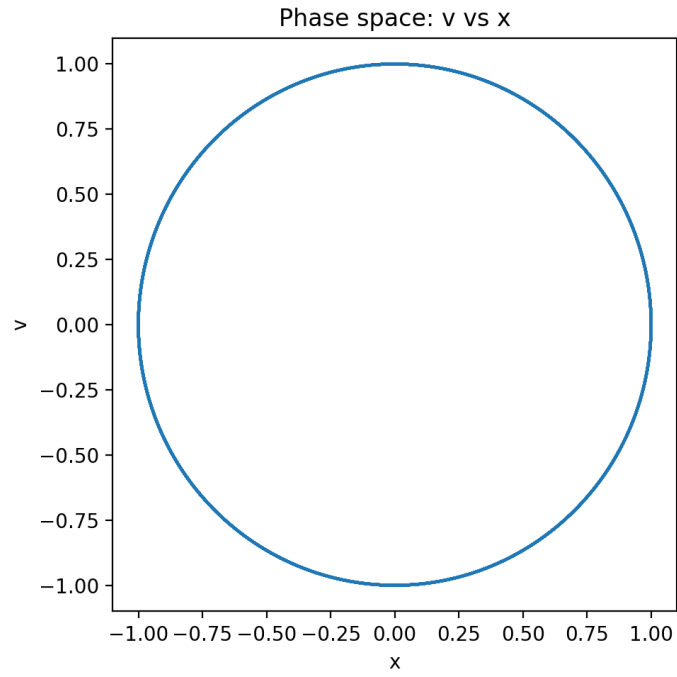


Figure 2: Phase-space trajectory $v(x)$ for the undamped oscillator ($\alpha = 0$).

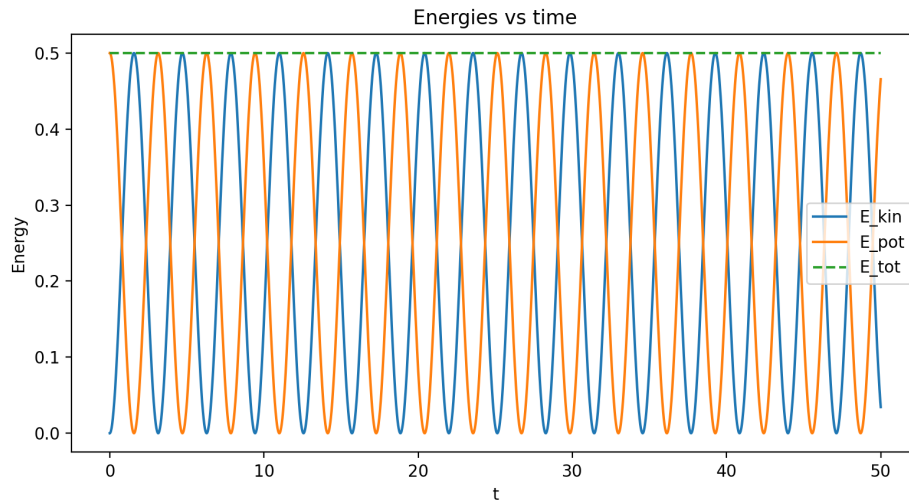


Figure 3: Kinetic, potential, and total energies for the undamped oscillator.

When a damping term was introduced ($\alpha = 0.1$), the numerical trajectory was compared with the analytical solution (Eq. 6), as shown in Figure 4. The numerical and exact curves overlap almost perfectly, confirming the correctness of the RK4 implementation. The phase-space trajectory (Figure 5) now spirals toward the origin, illustrating the gradual loss of energy due to friction. The corresponding energy plots (Figure 6) clearly show exponential decay of both kinetic and potential components.

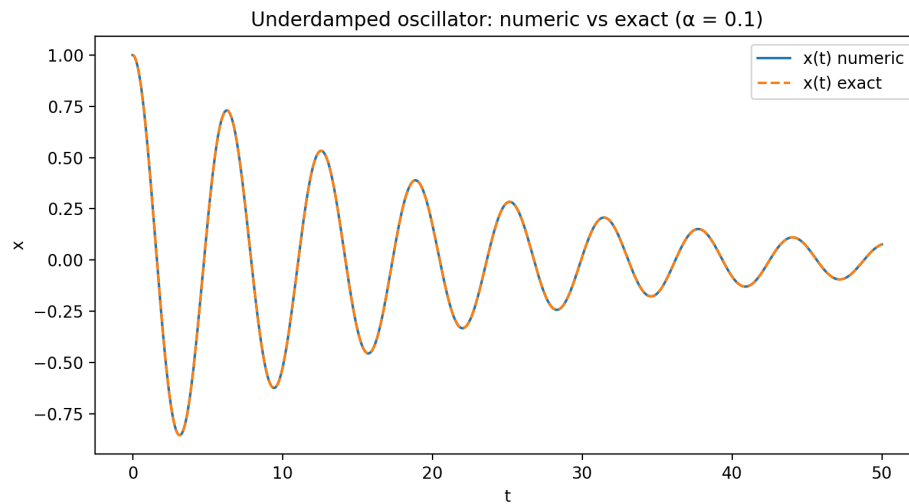


Figure 4: Comparison between numerical and analytical solutions for the underdamped case ($\alpha = 0.1$).

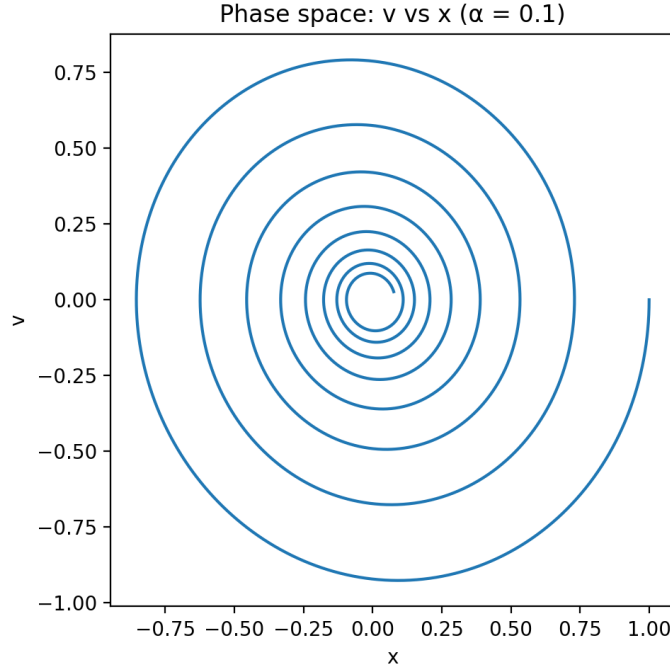


Figure 5: Phase-space trajectory $v(x)$ for the damped oscillator ($\alpha = 0.1$).

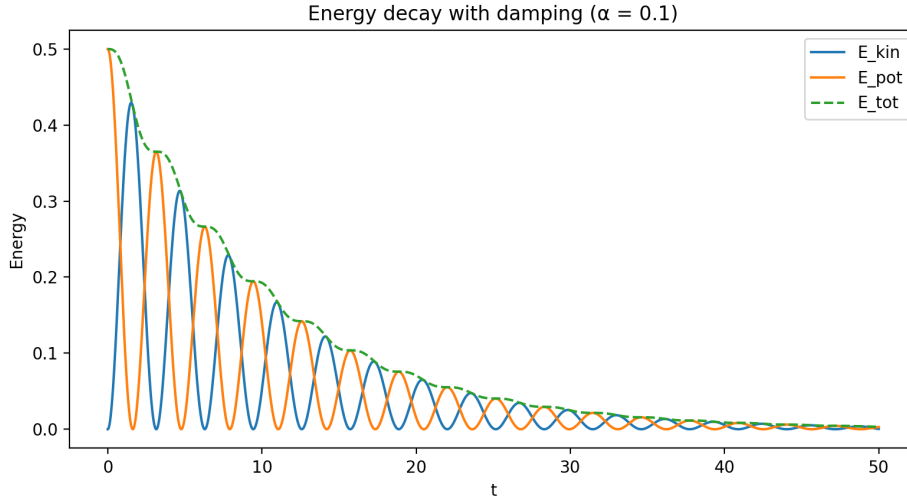


Figure 6: Energy decay in time for the damped oscillator ($\alpha = 0.1$).

The energy dissipation rate depends on the damping constant α . For small values of α , the amplitude decays slowly, and the energy remains significant over many oscillation periods. As α increases, the system loses energy more rapidly. Therefore, the dissipation is **fastest for large α** and **slowest for small α** , approaching zero in the ideal undamped case.

3.2 Energy Dissipation for Different Damping Coefficients

To study the influence of damping on the oscillator dynamics, simulations were performed for $\alpha = 10^{-4}, 0.1, 0.5$, and 1.95 with no external driving force ($F_{\text{ext}} = 0$). The resulting trajectories $x(t)$ are presented in Figure 7. As the damping coefficient increases, the amplitude of oscillations decreases more rapidly, and the system reaches equilibrium faster.

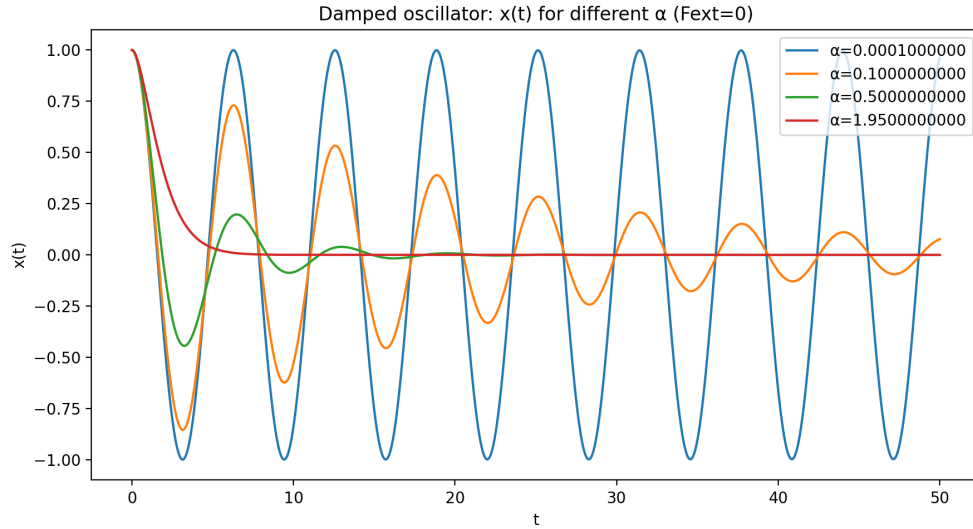


Figure 7: Time evolution of the oscillator position $x(t)$ for different damping coefficients α .

For very small damping ($\alpha = 10^{-4}$), the motion remains almost harmonic, with slowly decreasing amplitude. At moderate damping ($\alpha = 0.1$ and $\alpha = 0.5$), the oscillations decay exponentially but remain clearly visible. For $\alpha = 1.95$, which is close to the critical value $\alpha_{\text{crit}} = 2$, the amplitude quickly drops to zero and the oscillatory character nearly disappears.

In the **overdamped regime** ($\alpha > \alpha_{\text{crit}}$), the restoring force is insufficient to produce oscillations. Instead, the system slowly returns to equilibrium without crossing the equilibrium position. Thus, we expect a purely exponential relaxation toward $x = 0$, with no oscillatory behavior. This transition marks the boundary between underdamped and overdamped dynamics.

3.3 Driven Damped Oscillator and Mechanical Resonance

In the final part of the project, the oscillator was driven by an external periodic force $F_{\text{ext}}(t) = F_0 \sin(\Omega_{\text{ext}} t)$ with $F_0 = 1$ and damping parameter $\alpha = 1.0$. The first test simulation was performed for $\Omega_{\text{ext}} = 0.5 \omega_0$, as shown in Figure 8. After an initial transient period, the oscillations stabilized with a constant amplitude and frequency synchronized with the external driving force, confirming that the system reaches a steady-state regime dominated by the external excitation.

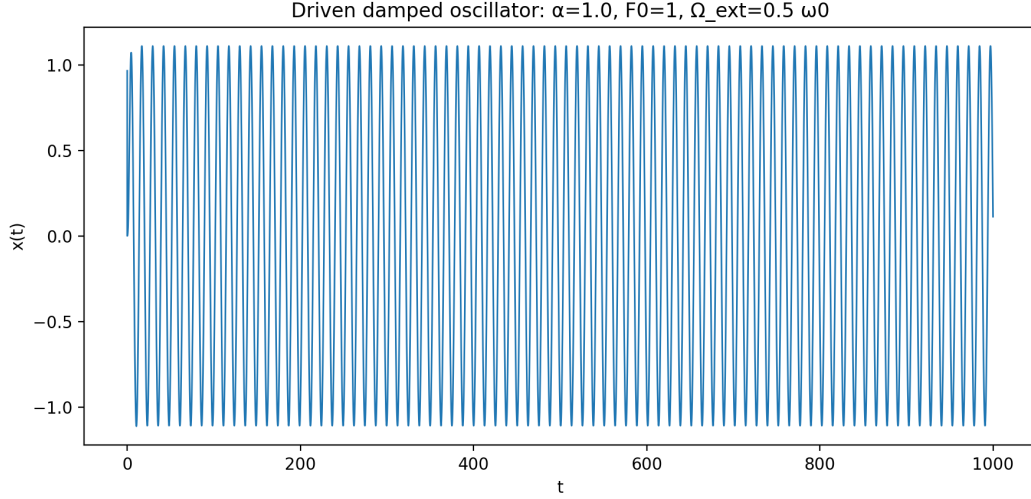


Figure 8: Driven damped oscillator ($\alpha = 1.0$, $F_0 = 1$) showing synchronization with external driving frequency $\Omega_{\text{ext}} = 0.5 \omega_0$.

To investigate the phenomenon of mechanical resonance, the driving frequency Ω_{ext} was varied in the range $0.1 \leq \Omega_{\text{ext}} \leq 2.0$, and the steady-state amplitudes x_{max} were measured for different damping values $\alpha = 0.01, 0.1, 0.5$, and 1.0 . The dependence of the driven amplitude on frequency is shown in Figure 9, along with the analytical curve for $\alpha = 1.0$ for comparison.

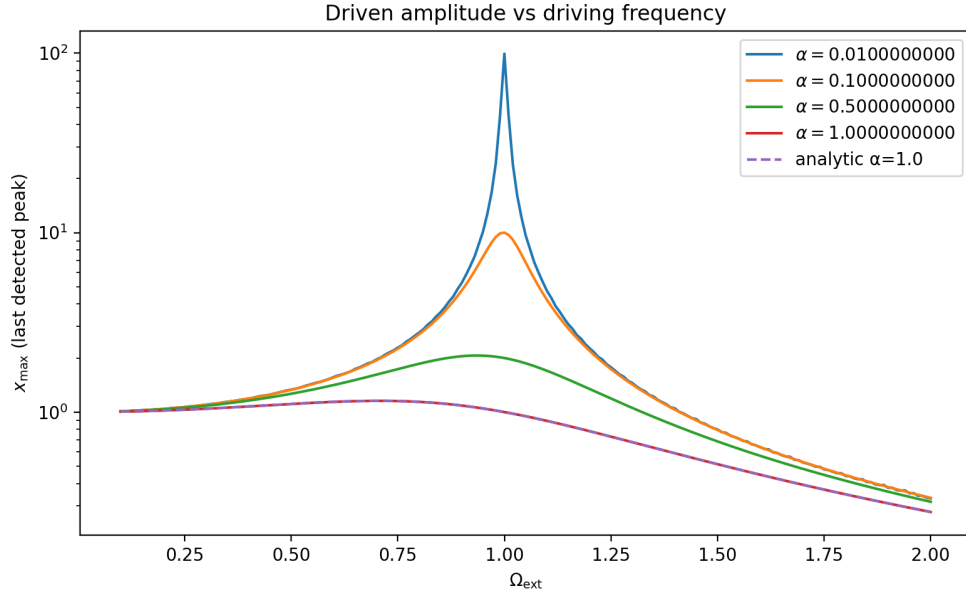


Figure 9: Driven oscillation amplitudes x_{max} as a function of the driving frequency Ω_{ext} for different damping coefficients. The dashed line shows the analytical result for $\alpha = 1.0$.

The results demonstrate a clear resonance peak centered near the natural frequency $\omega_0 = 1$. For small damping ($\alpha = 0.01$), the resonance is sharp and the amplitude grows rapidly as $\Omega_{\text{ext}} \rightarrow \omega_0$. With increasing damping, the resonance peak becomes broader and its maximum amplitude decreases significantly. This behavior agrees well with the analytical expression for the steady-state amplitude:

$$x_{\text{max}}(\Omega_{\text{ext}}) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \Omega_{\text{ext}}^2)^2 + (2\Omega_{\text{ext}}/\tau)^2}},$$

where $\tau = 2m/\alpha$ represents the damping timescale. Overall, the numerical results accurately reproduce the expected features of mechanical resonance in a damped driven harmonic oscillator.

4 Conclusions

The numerical simulations performed using the fourth-order Runge–Kutta method successfully reproduced the expected behavior of a one-dimensional harmonic oscillator under various conditions. For the undamped case, total energy was conserved and the motion remained purely periodic. Introducing damping led to exponential decay of energy and spiraling trajectories in phase space, consistent with analytical predictions. Increasing the damping coefficient shortened the relaxation time, and near the critical value the oscillations disappeared. When the system was driven by a periodic external force, resonance was observed near the natural frequency, with the amplitude and sharpness of the resonance peak decreasing with stronger damping. Overall, the results confirmed both the accuracy of the RK4 method and the theoretical understanding of damped and driven oscillatory motion.

References

- [1] Dr. Hab. Eng. Chwiej, T. *Project: Simulation of mechanical resonance for one-dimensional harmonic oscillator with RK4 method*. AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/2_harmonic_oscillator_RK4.pdf